Anticiper les départs pour réduire les risques internes

Lakeside®

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PROBLÈME

Coût du turnover

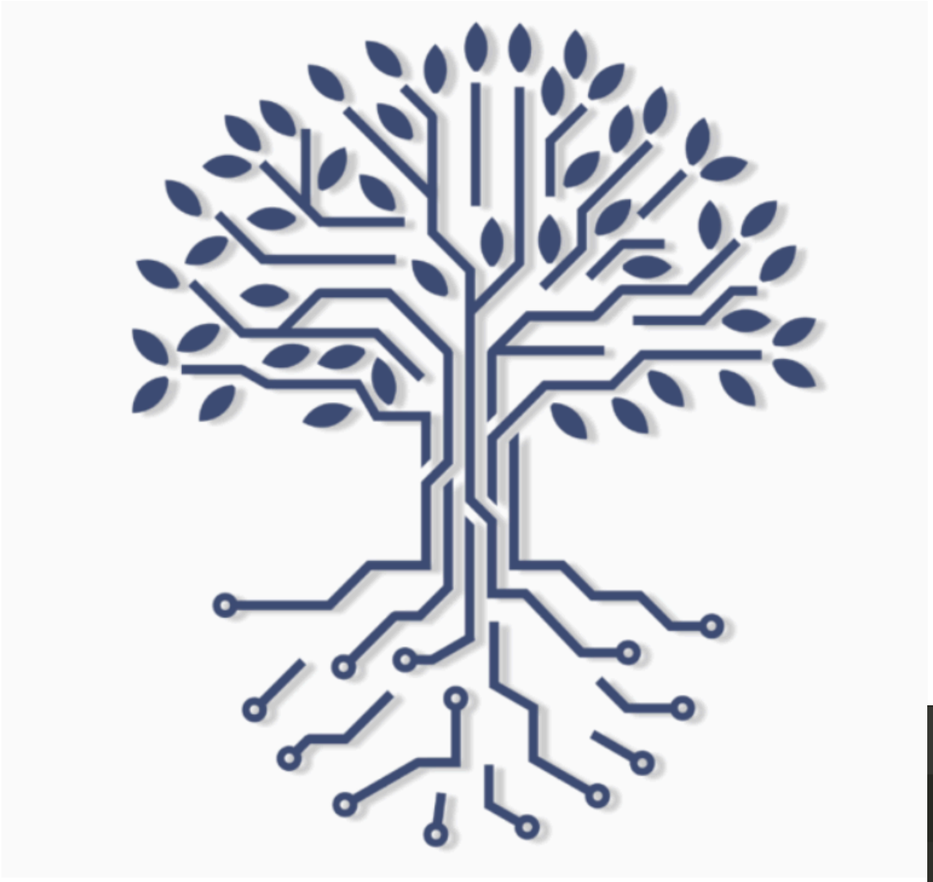
Perte de connaissance

Désorganisation

ANGLE CYBER

- **Anonymisation stricte** : Suppression totale des données identifiantes directes (Noms, Emails)
- **Pseudonymisation cryptographique :** Hachage des identifiants (SHA-256 avec "Sel") pour bloquer la rétro-ingénierie.
- **Séparation des Privilèges :** L'IA n'a accès qu'aux données chiffrées. Seule la DRH possède la clé de correspondance en local.

ANGLE FRUGALE



- **Optimisation du fichier .csv original**. Taille divisée par 2
- **Choix du modèle basé sur la consommation de gCO2**

```
Size before optimization : 336.78 KB
```

	Modèle	Accuracy	F1-Score	AUC	Émissions CO2 (mg)
0	Random Forest	0.809524	0.684211	0.795351	0.525983
1	Logistic Regression	0.777778	0.650000	0.816327	1.227675
2	Gradient Boosting	0.761905	0.615385	0.806689	0.613342

MODELE / SOLUTION

```python
models = {
    "Random Forest": RandomForestClassifier(n_estimators=100, random_state=42),
    "Logistic Regression": LogisticRegression(max_iter=1000, random_state=42),
    "Gradient Boosting": GradientBoostingClassifier(random_state=42)
}

comparison_results = []

for name, model in models.items():
    tracker = EmissionsTracker(project_name=name.replace(" ", "_"))
    current_pipeline = Pipeline(steps=[
        ('preprocessor', preprocess),
        ('classifier', model)
    ])

    tracker.start()
    current_pipeline.fit(X_train, y_train)
    emissions = tracker.stop()

    y_pred = current_pipeline.predict(X_test)
    y_proba = current_pipeline.predict_proba(X_test)[:, 1]

    comparison_results.append({
        "Modèle": name,
        "Accuracy": accuracy_score(y_test, y_pred),
        "F1-Score": f1_score(y_test, y_pred),
        "AUC": roc_auc_score(y_test, y_proba),
        "Émissions CO2 (mg)": emissions * 1000000
    })

df_compare = pd.DataFrame(comparison_results).sort_values(by="F1-Score", ascending=False)
display(df_compare)
```

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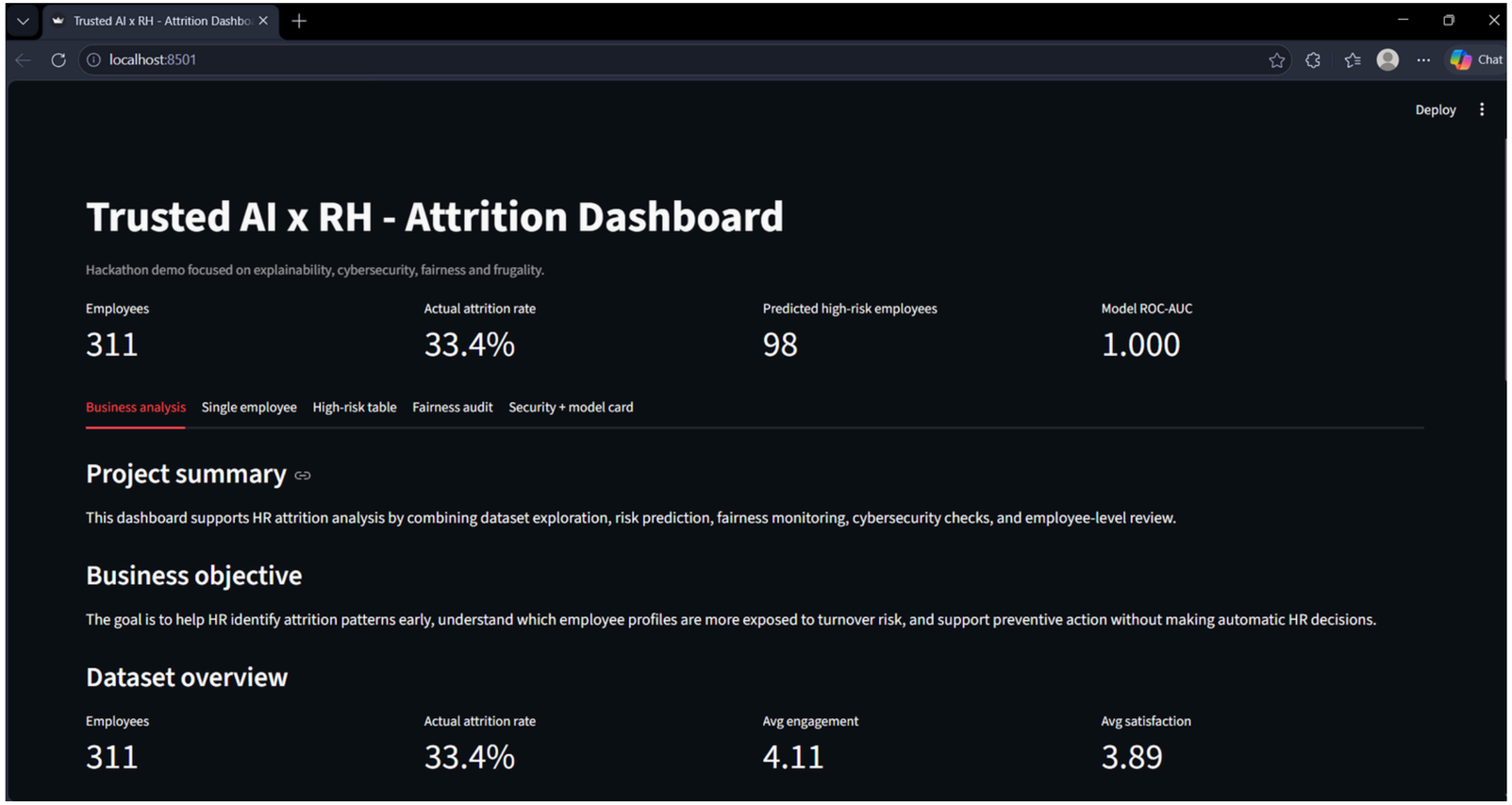
MODELE / SOLUTION

EmpID	Risque_%	Facteur_1	Facteur_2	Facteur_3
a405e94f2	92,5	num__TenureYears (14.093150684931507)	num__EmpSatisfaction (3)	num__DaysLateLast30 (4)
40a47ac1a	90,22	num__TenureYears (17.205479452054796)	num__EmpSatisfaction (3)	num__EngagementSurvey (5.0)
52465040	82,27	num__TenureYears (14.961643835616439)	num__ManagerID (22.0)	num__EmpSatisfaction (3)
c02db012f	81,72	num__TenureYears (15.668493150684931)	num__EmpSatisfaction (2)	num__DaysLateLast30 (3)
ffa5487c8a	78,32	num__TenureYears (15.556164383561644)	num__ManagerID (18.0)	num__EmpSatisfaction (3)
2158e1eb	75,78	num__TenureYears (18.202739726027396)	num__EmpSatisfaction (5)	num__ManagerID (16.0)
fb96e9d47	73,23	num__TenureYears (15.191780821917808)	num__EmpSatisfaction (4)	num__ManagerID (20.0)
a62ea4a3(	71,59	num__TenureYears (14.30958904109589)	num__EmpSatisfaction (4)	num__ManagerID (22.0)
4664e600	70,26	num__TenureYears (17.397260273972602)	num__EmpSatisfaction (2)	num__EngagementSurvey (4.5)
cf06f4404	69,53	num__TenureYears (13.715068493150685)	num__EmpSatisfaction (3)	num__EngagementSurvey (4.83)
e206d46ea	66,35	num__TenureYears (18.375342465753423)	num__EmpSatisfaction (3)	cat__RecruitmentSource_Google Search (Google Search)
f9ff24abdf	65,57	num__TenureYears (11.202739726027398)	num__EmpSatisfaction (5)	num__ManagerID (22.0)
9827490b	62,23	num__TenureYears (15.901369863013699)	num__ManagerID (19.0)	num__EmpSatisfaction (3)
c0c5d0a95	61,94	num__TenureYears (14.961643835616439)	num__EmpSatisfaction (5)	cat__RecruitmentSource_Google Search (Google Search)
0e584c65(	61,47	num__TenureYears (14.194520547945206)	num__EmpSatisfaction (5)	cat__RecruitmentSource_Google Search (Google Search)
f677548fe	61,47	num__TenureYears (10.202739726027398)	num__EmpSatisfaction (4)	num__EngagementSurvey (4.1)

DASHBOARD


Trusted AI x RH - Attrition Dashbo
localhost:8501
Chat
Deploy
Trusted AI x RH - Attrition Dashboard
Hackathon demo focused on explainability, cybersecurity, fairness and frugality.
Employees
311
Actual attrition rate
33.4%
Predicted high-risk employees
98
Model ROC-AUC
1.000
Business analysis
Single employee
High-risk table
Fairness audit
Security + model card
Project summary
This dashboard supports HR attrition analysis by combining dataset exploration, risk prediction, fairness monitoring, cybersecurity checks, and employee-level review.
Business objective
The goal is to help HR identify attrition patterns early, understand which employee profiles are more exposed to turnover risk, and support preventive action without making automatic HR decisions.
Dataset overview
Employees
311
Actual attrition rate
33.4%
Avg engagement
4.11
Avg satisfaction
3.89